

■ HERWATCH: WOMEN SAFETY ANALYTICS SYSTEM

A Real-Time Computer Vision & Geospatial Surveillance Framework for Enhanced Public Safety

Executive Summary: HerWatch is an end-to-end intelligent safety surveillance system combining state-of-the-art computer vision (YOLOv8, MediaPipe, MobileNetV2) and geospatial analytics (Folium, Geopy) to protect women in urban and semi-urban environments. The system delivers real-time distress signal detection, automated nocturnal lone-woman identification, contextual crowd gender monitoring, and interactive crime hotspot visualization over a responsive React dashboard.

Project Version: 1.0.0 Production	Tech Stack: Python (Flask), React.js, PyTorch, OpenCV
Target Domain: Public Safety & Smart Cities	AI Architecture: YOLOv8 + MediaPipe + MobileNetV2
Document Type: End-to-End System Report	Generated Date: August 2026

1. Problem Statement & Strategic Vision

Women's safety in public spaces remains a critical global challenge, particularly during late hours or in isolated transit areas. Traditional CCTV networks are inherently passive—requiring continuous human monitoring which is prone to fatigue, latency, and delayed emergency response. Furthermore, conventional security systems lack situational awareness regarding distress gestures, gender dynamics, and localized crime historical data.

HerWatch addresses these shortcomings by introducing an proactive, AI-driven surveillance framework that automatically monitors live video streams and recorded feeds to identify high-risk scenarios in real-time. By combining spatial crime intelligence with multi-modal deep learning models, HerWatch empowers municipal authorities, campus security, and law enforcement with actionable alerts.

Key Strategic Pillars:

- **Proactive Distress Detection:** Instant recognition of non-verbal SOS gestures (hand waving, double hand raise, cover mouth) without physical button pressing.
- **Nocturnal Contextual Risk Assessment:** Detecting solitary women during high-risk nighttime hours (19:00 - 06:00) with rapid male/female crowd ratio analysis.
- **Geospatial Crime Intelligence:** Dynamic crime hotspot mapping and proximity calculations against historical district-level Indian crime datasets.

- **Dual-Mode Accessibility:** Supporting both low-latency live camera streaming (WebSocket) and file-based recorded video processing.

2. System Architecture & High-Level Design

HerWatch employs a decoupled client-server architecture. The backend is powered by Flask and OpenCV, executing heavy AI pipeline processing. The frontend is built with React.js, providing real-time dashboard analytics, WebSocket video feeds, and interactive map displays.

Component Layer	Technologies Used	Key Responsibilities
Frontend UI Layer	React 18, Material-UI, Framer Motion, Chart.js	Live alert monitoring, interactive hotspot maps, video upload UI, gesture status telemetry.
API & WebSocket Backend	Flask 3.1, Flask-CORS, Flask-Sock (WebSocket)	RESTful endpoints (<code>/api/hotspots/analyze`</code> ), WebSocket camera feed handlers ( <code>/ws/camera`</code>), static serving.
Object Detection Engine	YOLOv8 (Ultralytics), PyTorch, OpenCV	Real-time person bounding box detection (<code>^class=0`</code> ), confidence thresholding ( <code>>0.4`</code>), tracking.
Gesture Detection Pipeline	MediaPipe Holistic Solution (3D Keypoints)	Hand & body pose landmark tracking; wave frequency analysis, hand-above-wrist elevation logic.
Gender Classification Engine	MobileNetV2 (PyTorch Torchvision Weights)	Face region cropping from person bboxes, image normalization, gender probability inference.
Geospatial Engine	Folium, Leaflet.js, Geopy (Nominatim & Geodesic)	District geocoding, geodesic distance calculation (<code><100km`</code> radius), interactive risk circle rendering.

3. Core AI Modules & Deep Learning Pipeline

3.1 SOS Distress Gesture Detection Engine

The distress gesture module utilizes **MediaPipe Holistic** to extract 33 pose landmarks and 21 hand keypoints per hand in real-time. It runs heuristic geometric evaluation on body coordinates without requiring heavy custom gesture training models, ensuring sub-50ms latency.

Detection Logic & Mathematical Conditions:

- **Raised Hand Gesture:** Evaluates if $\text{Middle_Finger_Tip.y} < \text{Wrist.y}$ and $\text{Wrist.y} < 0.5$ (upper frame half).
- **Hand Waving Gesture:** Tracks horizontal displacement $|\text{Middle_Tip.x} - \text{Wrist.x}| > 0.15$ combined with vertical displacement > 0.25 over a sliding 2.0-second time window.
- **Double Hand SOS:** Validates both left and right hand keypoints simultaneously elevated above shoulder line ($\text{Left_Wrist.y} < \text{Left_Shoulder.y}$ & $\text{Right_Wrist.y} < \text{Right_Shoulder.y}$).
- **Hand Over Mouth Sign:** Calculates Euclidean distance between **Hand_Landmarks** and **Mouth_Landmark** (threshold < 0.15 normalized units).

3.2 Lone Woman Detection at Night & Gender Analytics

Solitary exposure during late hours represents one of the highest statistical risk scenarios for women in urban transit zones. HerWatch combines temporal monitoring with person detection and deep-learning gender classification.

Processing Pipeline Breakdown:

1. **Nocturnal Time Filter:** Checks system timestamp; triggers active lone-woman assessment if local time falls between 19:00 (7 PM) and 06:00 (6 AM).
2. **Person Detection (YOLOv8n):** Scans image frame for objects with COCO class ID 0 (person). Detections below 0.40 confidence are filtered out.
3. **Face / Body Extraction:** Crops upper 60% of bounding box (representing head and torso) for gender classification.
4. **MobileNetV2 Inference:** Resizes crop to 224x224 RGB, normalizes tensor values, and feeds into MobileNetV2. Softmax activation predicts Female vs Male confidence score.
5. **Lone Woman Rule Evaluation:** If $\text{Total_Persons} == 1$, $\text{Gender} == \text{Female}$, $\text{Confidence} \geq 0.60$, and $\text{is_nighttime}() == \text{True}$, an emergency **LONE WOMAN AT NIGHT** alert is dispatched.

3.3 Geospatial Crime Hotspot Mapping & Geocoding

HerWatch incorporates district-level historical crime dataset (women-crimesdataset-India.csv) covering multiple crime categories including Rape, Kidnapping & Abduction, Dowry Deaths, Assault on Women, and Cruelty by Husband/Relatives.

Geospatial Distance & Risk Assessment Algorithm:

- **Geocoding with Nominatim:** User inputs city/district name (e.g., 'Bengaluru'). Nominatim API returns exact (latitude, longitude) coordinates with automatic retry logic.

- **Geodesic Distance Computation:** Computes great-circle distance between user location (P_{user}) and every district $(P_{district})$ using Geopy's geodesic formula:
Distance Formula: $d = 2R \cdot \arcsin(\sqrt{\sin^2(\Delta lat/2) + \cos(lat1) \cdot \cos(lat2) \cdot \sin^2(\Delta lon/2)})$
- **Proximity Filter:** Filters all crime records within a **100 km radius**.
- **Risk Categorization:** Assigns risk levels based on total crimes reported: High Risk (Red Circle, >1000 incidents), Medium Risk (Orange Circle), Low Risk (Green Circle).

4. Backend Architecture & REST/WebSocket API Reference

The Flask backend server (`app.py`) runs on port 5001 and serves both RESTful endpoints and high-performance WebSockets for live video streaming.

HTTP/WS Method	Endpoint Route	Description & Output Schema
GET	/api/health	Health check endpoint. Returns <code>{"status": "ok", "message": "API is running"}</code> .
POST	/api/hotspots/analyze	Accepts location name or lat/lon coordinates. Computes nearby crime hotspots within 100km and returns Folium interactive map HTML string.
POST	/analyze_video	Uploads video file (<code>multipart/form-data</code>). Runs frame-by-frame YOLOv8 + MediaPipe analysis and generates processed output video file.
GET	/api/live-camera/list	Enumerates connected physical webcams/cameras using OpenCV device index probing (<code>cv2.VideoCapture(i)</code>).
WebSocket	/ws/camera	Bidirectional socket. Receives Base64 encoded JPEG video frames from frontend; sends processed frame Base64 with alert telemetry JSON payload.

5. Frontend Application Architecture (React.js)

The React frontend (`sos-detection-frontend`) provides a modern, responsive UI crafted with Material-UI (MUI v5) and Framer Motion animations. It features dynamic tab routing for seamless navigation across all core functions.

Frontend Page Breakdown:

Page Component	File Path	Key Features & User Interactions
Live Camera Stream	src/pages/LiveCamera.js	Selects active webcam, opens WebSocket stream to Flask, renders annotated live video canvas, triggers auditory alert sounds, displays active threat banner.
Video Analysis Mode	src/pages/VideoMode.js	Drag-and-drop video uploader (`.mp4`, `.avi`), progress indicator bar, frame playback, detected threats timeline log.
Hotspot Risk Map	src/pages/HotspotMap.js	Location search input, 'Use My Current GPS Location' button, embedded Folium Leaflet interactive heatmap, crime statistic breakdown cards.
Analytics Dashboard	src/pages/Dashboard.js	Chart.js visual charts displaying total incident breakdown (Rape, Assault, Kidnapping) and hourly risk distribution.

6. Installation, Configuration & Deployment Guide

Prerequisites:

- Python 3.10+ (Virtual environment recommended)
- Node.js 18+ and npm 9+
- Webcam device (for live streaming mode)

Step 1: Backend Setup & Execution

```
cd "/Users/spurthinaveli/Desktop/Herwatch/ThemeBased Code"
source venv/bin/activate
# Install dependencies if setting up fresh:
pip install -r requirements.txt
pip install "numpy<2" reportlab
python app.py
```

Step 2: Frontend Setup & Execution

```
cd "/Users/spurthinaveli/Desktop/Herwatch/ThemeBased Code/sos-detection-frontend"
# Start React Development Server:
npm start
```

7. Future Scope & System Enhancements

- **Audio Distress Recognition:** Integrating micro-acoustic neural networks (such as YAMNet or PyAudio) to detect screams, glass breaking, and vocal distress cues in ambient audio feeds.
 - **Edge Computing Deployment:** Quantizing YOLOv8 and MobileNetV2 models for TensorRT / ONNX runtime to deploy on edge AI hardware (Nvidia Jetson Orin Nano, Raspberry Pi 5).
 - **Automated Emergency Dispatch:** Integrating Twilio API to automatically dispatch real-time SMS alerts with live GPS pin coordinates to designated emergency contacts and local police authorities.
 - **Infrared / Thermal Camera Support:** Fine-tuning person and pose detection pipelines on thermal and night-vision video channels for zero-light environment detection.
-

End of HerWatch Technical Report — Generated automatically for HerWatch Women Safety Project.